

# ABSTRACT

Shiro Abhyanga is a traditional Ayurvedic head massage therapy widely practiced for promoting mental relaxation, stress relief, neurological balance, and overall well-being. The therapeutic effectiveness of Shiro Abhyanga mainly depends on proper massage execution, rhythmic hand movement, pressure consistency, posture alignment, and suitable oil application according to the individual's Prakriti. In conventional Ayurvedic practice, evaluation of massage quality is primarily based on practitioner observation and personal expertise, which introduces subjectivity and lacks standardized assessment methods. The absence of objective evaluation techniques makes it difficult to ensure consistency in therapist training, quality assurance, and performance monitoring.

To address these limitations, this project proposes an integrated Internet of Things (IoT) and Artificial Intelligence (AI) based framework for objective assessment and monitoring of Shiro Abhyanga massage therapy. The proposed system combines both hardware and software components to create a smart healthcare solution capable of analysing massage posture, movement smoothness, physiological conditions, and therapy execution quality in real time.

The hardware subsystem consists of wearable and environmental monitoring modules developed using the ESP32 microcontroller integrated with multiple sensors such as pulse sensor, temperature sensor, humidity sensor, force-sensitive sensor, and MPU6050 motion sensor. These sensors continuously monitor physiological and environmental parameters during massage sessions and transmit the collected data to a cloud-based dashboard through Wi-Fi connectivity for real-time visualization and analysis.

The software subsystem integrates machine learning, deep learning, and computer vision techniques for intelligent posture evaluation and movement analysis. Prakriti prediction is performed using supervised machine learning algorithms trained on questionnairebased Ayurvedic datasets. Based on the predicted Prakriti type, suitable medicated oil recommendations are generated using Ayurvedic rule-based mapping.

For posture evaluation, real-time massage videos are analysed using feature-based machine learning models, YOLO-based deep learning posture detection, biomechanical angle analysis, and optical flow motion analysis. Numerical features such as hand

coordinates, velocity, acceleration, jerk, motion smoothness, and hand-to-head distance are extracted from video frames and stored in CSV format for supervised learning.

Experimental results demonstrate that the proposed framework successfully combines IoT monitoring with AI-based posture evaluation to provide an explainable, non-invasive, and real-time assessment mechanism for Shiro Abhyanga therapy. The developed system contributes toward standardization of Ayurvedic massage evaluation, therapist training support, quality assurance, and modernization of traditional healthcare practices using intelligent computational technologies.

**Keywords:** Shiro Abhyanga, Ayurveda, Internet of Things, Machine Learning, Deep Learning, Computer Vision, YOLO, ESP32, Posture Analysis, Optical Flow, Biomechanical Analysis, Prakriti Prediction.

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# Chapter 1

## Introduction

### 1.1 Introduction

Ayurveda is one of the oldest traditional healthcare systems and includes several therapeutic massage procedures intended to improve physical and mental well-being. Among these therapies, Shiro Abhyanga is a widely practiced Ayurvedic head massage therapy used to reduce stress, improve blood circulation, and promote neurological relaxation. The success of this therapy depends on proper massage posture, rhythmic hand movement, correct pressure application, and patient-specific oil selection. In conventional practice, massage quality is evaluated mainly through therapist experience and visual observation. Such assessment methods are subjective and vary between practitioners, making standardization difficult. In addition, traditional systems lack continuous physiological monitoring and real-time feedback mechanisms that can assist therapists during therapy sessions.

Recent advancements in Internet of Things (IoT), machine learning, and computer vision technologies provide opportunities for developing intelligent healthcare systems capable of objective movement analysis and real-time monitoring. IoT sensors enable collection of physiological and environmental data, while Artificial Intelligence (AI) based models can analyse posture correctness, movement smoothness, and motion consistency from video streams. This project introduces a complete hardware and software framework that integrates wearable IoT monitoring devices with machine learning and deep learning models for evaluating Shiro Abhyanga massage quality. The system combines sensor-based monitoring, cloud dashboards, posture classification, YOLO-based detection, biomechanical analysis, optical flow analysis, and Ayurvedic personalization using Prakriti prediction.

The hardware subsystem includes ESP32 microcontroller, pulse sensor, temperature and humidity sensors, force-sensitive sensors, and motion sensors for real-time monitoring during therapy sessions. The collected sensor data is transmitted to a cloud-based dashboard for visualization and analysis. The software subsystem uses feature-based machine learning, deep learning models, and computer vision techniques to analyse massage posture and motion dynamics. The system classifies massage execution as proper or improper based on posture consistency, movement smoothness, and ergonomic alignment. In addition, Prakriti prediction and Taila recommendation modules provide Ayurvedic personalization according to traditional healthcare principles. The proposed framework aims to support therapist training, improve therapy quality assurance, and modernize traditional Ayurvedic therapies using intelligent computational methods.

## Chapter 2

# Literature Survey

## 2.1 Literature Survey

Shiro Abhyanga and other Ayurvedic massage therapies have been traditionally evaluated based on practitioner experience and visual observation. While several studies have established the therapeutic benefits of head massage and oil-based therapies, there is limited work focusing on the objective assessment of stroke quality using computational methods. Recent advances in computer vision, motion analysis, and machine learning have enabled quantitative evaluation of human movement, which can be applied to massage-based therapies for standardized assessment.

- Nikam et al. reviewed classical Ayurvedic literature on Shiro Abhyanga and reported its effectiveness in reducing stress, improving sleep quality, and supporting neurological balance. The study emphasized the importance of proper stroke execution but relied entirely on qualitative descriptions without measurable indicators.
- Chandravanshi et al. conducted a clinical study on head massage therapies and observed improvements in relaxation and circulation. Although therapeutic benefits were confirmed, the assessment of massage technique remained subjective and practitioner dependent.
- Jia et al. proposed a deep learning framework for recognizing massage gestures using video data. Their work demonstrated that convolutional neural networks can successfully classify massage actions; however, the study focused on gesture recognition rather than quality assessment or smoothness analysis.
- Liu et al. developed a wearable-sensor-based deep learning system for real-time massage action recognition. While effective in capturing motion patterns, the system required additional hardware and did not analyze biomechanical posture or stroke continuity from video data.
- Farneback introduced a dense optical flow method for estimating frame-to-frame motion, which has since been widely used for smoothness analysis and motion stability assessment in human movement studies. This technique provides a foundation for quantifying stroke continuity in massage applications.
- Horn and Schunck presented an early formulation of optical flow for motion estimation, establishing fundamental principles for analyzing temporal movement patterns in image sequences.
- Tripathi et al. reviewed motion analysis techniques used for evaluating therapeutic massage and physiotherapy exercises. The authors highlighted the lack of automated systems for assessing massage quality and emphasized the need for objective metrics.

- Cao et al. introduced OpenPose for real-time multi-person pose estimation. Although pose-estimation models enable accurate joint tracking, their direct application to Ayurvedic massage quality assessment remains underexplored.
- Breiman proposed the Random Forest algorithm, which has been widely used for classification tasks involving structured features. Its robustness to noise makes it suitable for biomechanical and motion-derived feature sets.
- Chen and Guestrin introduced XGBoost, a scalable gradient boosting framework that has demonstrated strong performance in structured-data classification, including human activity recognition.
- Goodfellow et al. discussed deep learning architectures for visual and temporal data analysis, providing foundational concepts for CNN-based movement classification.
- Dalal and Triggs proposed Histograms of Oriented Gradients (HOG) for human detection, influencing subsequent feature-extraction techniques for motion analysis.
- Zach et al. developed the TV-L1 optical flow algorithm, which improves robustness to noise and illumination changes and has been applied in smoothness estimation.
- Singh et al. reviewed applications of machine learning in traditional medicine and emphasized the potential of data-driven approaches for standardizing manual therapies.
- Patwardhan et al. explored Prakriti classification and its role in personalized Ayurvedic treatment. Their work established the relevance of constitution analysis for individualized therapy planning.

## 2.2 Research Gap

The literature survey indicates that while the therapeutic benefits of Shiro Abhyanga and related massage techniques are well documented, objective assessment of stroke quality remains largely unexplored. Existing Ayurvedic studies rely on qualitative evaluation, whereas computer vision and machine learning research primarily focuses on gesture recognition or activity classification without addressing massage quality or smoothness.

Furthermore, most motion-analysis systems either depend on wearable sensors or generic pose-estimation frameworks, which are not tailored to the specific requirements of Ayurvedic massage evaluation. There is also a lack of integrated approaches that combine biomechanical analysis, motion smoothness, statistical feature modeling, and personalized Ayurvedic context.

The proposed system addresses these gaps by introducing a multi-pipeline framework for objective Shiro Abhyanga stroke assessment using video-based biomechanical features, optical-flow-derived smoothness metrics, machine learning classification, and Prakriti-based personalization. This approach contributes toward standardized evaluation, improved training, and data-driven support for traditional Ayurvedic practices.

# Chapter 3

## Methodology

### 3.1 Overall System Architecture

The proposed system combines both hardware and software components to create an intelligent framework for monitoring and evaluating Shiro Abhyanga massage therapy. The system integrates Internet of Things (IoT) based wearable monitoring devices with Artificial Intelligence (AI), machine learning, deep learning, and computer vision techniques for objective posture assessment and real-time healthcare monitoring. The complete framework is divided into two major subsystems:

1. Hardware-Based Monitoring System
2. Software-Based Posture Evaluation System

The hardware subsystem continuously collects physiological and environmental parameters during therapy sessions using multiple sensors connected to the ESP32 microcontroller. The collected data is transmitted to a cloud dashboard through Wi-Fi connectivity for real-time monitoring and visualization.

The software subsystem processes real-time massage videos using machine learning and deep learning models. The system analyses hand posture, movement smoothness, motion consistency, and ergonomic alignment to classify massage execution as proper or improper.

### 3.2 Hardware Methodology

The hardware module is designed as a wearable and environmental monitoring system capable of collecting physiological and therapy-related data during Shiro Abhyanga sessions.

#### 3.2.1 Components Used

The major hardware components used in the proposed system are listed below:

- ESP32 Microcontroller
- Pulse Sensor
- Temperature Sensor
- Humidity Sensor



- Force Sensitive Sensor
- MPU6050 Motion Sensor
- OLED Display
- Rechargeable Battery
- Wi-Fi Connectivity Module

### **3.2.2 Working Principle**

The sensors continuously monitor physiological and environmental conditions during massage therapy. The pulse sensor measures heart rate variations, while the temperature and humidity sensors monitor surrounding environmental conditions. The force sensor analyses pressure consistency during massage execution, and the MPU6050 motion sensor captures movement stability and orientation.

All sensor readings are processed by the ESP32 microcontroller and transmitted to a cloud-based dashboard using Wi-Fi connectivity. The dashboard displays real-time values for monitoring and analysis purposes.

## **3.3 Software Methodology**

The software subsystem integrates machine learning, deep learning, and computer vision techniques for posture evaluation and Ayurvedic personalization.

### **3.3.1 Prakriti Prediction**

Prakriti prediction is performed using supervised machine learning algorithms trained on questionnaire-based Ayurvedic datasets. The system predicts the user's Prakriti type based on physical and physiological attributes.

Machine learning models used include:

- Logistic Regression
- Random Forest
- Gradient Boosting
- XGBoost

Based on the predicted Prakriti, the system recommends suitable medicated oils according to Ayurvedic principles.

### **3.3.2 Real-Time Video Acquisition**

Massage sessions are recorded using cameras under controlled lighting conditions. The videos are normalized in terms of frame rate and resolution before further analysis.

### **3.3.3 CSV-Based Feature Extraction**

Numerical features are extracted from massage videos and stored in CSV format. The extracted features include:

- Hand Coordinates
- Head Coordinates
- Velocity
- Acceleration
- Motion Entropy
- Jerk
- Hand-to-Head Distance

These features are used for supervised posture classification.

### **3.3.4 Machine Learning Based Posture Classification**

The extracted feature vectors are used to train machine learning models for posture correctness evaluation. The models classify massage execution as proper or improper based on movement consistency and posture alignment.

The models used include:

- Support Vector Machine (SVM)
- Random Forest
- Gradient Boosting
- XGBoost

### **3.3.5 YOLO-Based Deep Learning Detection**

A YOLO object detection model is trained using massage video frames to detect hand and arm positions in real time. The model generates bounding boxes and confidence scores for posture evaluation.

YOLO provides efficient real-time posture localization suitable for live therapy monitoring applications.

### **3.3.6 Biomechanical Angle Analysis**

Biomechanical angle analysis is performed to evaluate ergonomic posture alignment during massage execution. Joint angles such as elbow angle and wrist tilt are calculated using geometric relationships between limb coordinates.

The analysis helps identify posture deviations and movement inconsistencies.

### **3.3.7 Optical Flow Motion Analysis**

Optical flow analysis is used to evaluate movement smoothness and rhythmic consistency during massage sessions. Motion vectors are calculated between consecutive video frames to analyse motion stability and abrupt movements.

Improper massage execution generally exhibits higher jerk values and unstable motion patterns.

## **3.4 Cloud Dashboard Integration**

All hardware and software outputs are integrated into a centralized cloud dashboard for visualization and monitoring. The dashboard displays:

- Heart Rate
- Temperature
- Humidity
- Pressure Readings
- Motion Stability
- Posture Prediction
- Prakriti Prediction
- Taila Recommendation

The dashboard enables therapists and researchers to monitor therapy sessions in real time and analyse posture correctness efficiently.

# **Chapter 4**

# Results

## 4.1 Prakriti Prediction Results

Prakriti prediction was performed using multiple supervised machine learning models trained on questionnaire-based Ayurvedic datasets. The performance of different models was evaluated using classification accuracy.

| Model               | Accuracy (%) |
|---------------------|--------------|
| Logistic Regression | 89.00        |
| Random Forest       | 91.00        |
| XGBoost             | 93.00        |
| Gradient Boosting   | 94.50        |

Table 4.1: Performance of Prakriti Prediction Models

Among all evaluated models, Gradient Boosting achieved the highest accuracy of 94.50%, making it the best-performing model for Prakriti classification. The predicted Prakriti type was further used for personalized Ayurvedic oil recommendation.

## 4.2 CSV-Based Posture Prediction Results

Feature vectors extracted from massage videos were used to train machine learning models for posture correctness evaluation.

| Metric     | Value   |
|------------|---------|
| Accuracy   | 75.27%  |
| F1-Score   | 0.82    |
| Model Used | XGBoost |

Table 4.2: CSV-Based Posture Prediction Results

The results demonstrate that the extracted numerical features successfully differentiate proper and improper massage execution patterns.

## 4.3 Machine Learning and Deep Learning Results

Different machine learning and deep learning models were trained and evaluated for posture classification.

### 4.3.1 Machine Learning Model Performance

| Model               | Accuracy (%) |
|---------------------|--------------|
| Logistic Regression | 82.12        |
| Random Forest       | 85.06        |
| K-Nearest Neighbour | 74.70        |

|                        |       |
|------------------------|-------|
| Support Vector Machine | 89.11 |
| Gradient Boosting      | 83.82 |

Table 4.3: Machine Learning Model Performance

Support Vector Machine achieved the highest accuracy among machine learning models.

### 4.3.2 Deep Learning Model Performance

| Deep Learning Model | Accuracy (%) |
|---------------------|--------------|
| Simple CNN          | 89.11        |
| Deep CNN            | 89.11        |
| MobileNetV2         | 88.12        |
| VGG16               | 84.42        |

Table 4.4: Deep Learning Model Performance

The Deep CNN model provided high classification accuracy and improved visual feature extraction for posture analysis.

## 4.4 YOLO-Based Detection Results

YOLO object detection was used for real-time hand and arm posture localization during massage sessions.

| Metric                   | Value |
|--------------------------|-------|
| Detection Accuracy       | 92%   |
| Average Confidence Score | 0.85  |
| Real-Time Inference      | Yes   |

Table 4.5: YOLO Detection Results

The YOLO model successfully detected massage hand posture in real time with high confidence scores and accurate localization.

## 4.5 Biomechanical Angle Analysis Results

Biomechanical analysis was performed using elbow angles, wrist tilt, and posture symmetry measurements.

| Model                  | Accuracy |
|------------------------|----------|
| Random Forest          | 94.04%   |
| Support Vector Machine | 88.02%   |

Table 4.6: Biomechanical Angle Classification Results

Random Forest achieved the highest accuracy for biomechanical posture analysis due to its capability to model non-linear feature relationships effectively.

## **4.6 Optical Flow Motion Analysis Results**

Optical flow analysis was used to evaluate movement smoothness and rhythmic consistency during massage sessions.

Proper massage execution showed:

- Stable motion patterns
- Lower jerk values
- Consistent movement rhythm
- Smooth hand transitions

Improper massage execution showed:

- Abrupt movements
- Higher jerk values
- Irregular motion flow
- Unstable movement patterns

The optical flow analysis successfully differentiated proper and improper massage execution based on motion smoothness characteristics.

## **4.7 IoT Hardware Monitoring Results**

The IoT hardware subsystem successfully monitored physiological and environmental parameters during therapy sessions.

The system continuously monitored:

- Heart Rate
- Temperature
- Humidity
- Pressure Consistency
- Motion Stability

The ESP32 microcontroller successfully transmitted sensor readings to the cloud dashboard in real time without communication interruption.

## 4.8 Discussion

The obtained results demonstrate that integrating IoT monitoring with machine learning and deep learning techniques provides an effective framework for objective evaluation of Shiro Abhyanga massage therapy.

Machine learning and deep learning models successfully classified massage posture using extracted motion features and visual analysis. YOLO-based posture detection enabled efficient real-time localization of hand movements, while biomechanical angle analysis and optical flow methods provided additional validation for posture correctness and movement smoothness.

The IoT subsystem enhanced the framework by enabling real-time physiological and environmental monitoring during therapy sessions. The cloud dashboard provided centralized visualization of both hardware and software outputs.

Overall, the proposed system demonstrates the successful integration of Ayurveda, Artificial Intelligence, Internet of Things, and computer vision technologies for intelligent healthcare monitoring and therapy assessment.

# Chapter 5

## Conclusion

### 5.1 Conclusion

This project successfully developed an integrated Internet of Things (IoT) and Artificial Intelligence based framework for objective evaluation and monitoring of Shiro Abhyanga massage therapy. The proposed system combines wearable healthcare monitoring devices with machine learning, deep learning, and computer vision techniques to create a smart Ayurvedic healthcare solution.

The hardware subsystem continuously monitored physiological and environmental parameters using sensors such as pulse sensor, temperature sensor, humidity sensor, forcesensitive sensor, and MPU6050 motion sensor connected to the ESP32 microcontroller. The collected data was transmitted to a cloud dashboard for real-time monitoring and visualization.

The software subsystem analysed massage posture and movement dynamics using feature-based machine learning models, YOLO object detection, biomechanical angle analysis, and optical flow motion analysis. The system effectively classified massage execution as proper or improper based on posture alignment, movement smoothness, and motion consistency.

Prakriti prediction and Ayurvedic Taila recommendation modules provided personalized healthcare support according to Ayurvedic principles. Experimental results demonstrated that the proposed framework achieved high accuracy in posture detection, motion analysis, and classification tasks.

The project successfully establishes a connection between traditional Ayurvedic therapies and modern computational technologies such as Artificial Intelligence, IoT, and computer vision. The proposed framework can support therapist training, improve therapy quality assurance, and contribute to the modernization of Ayurvedic healthcare systems.

## **5.2 Future Scope**

Although the proposed system achieved promising results, several improvements and extensions can be implemented in future work.

### **5.2.1 Real-Time Feedback System**

Future systems can provide instant corrective feedback to therapists during massage sessions using real-time audio or visual guidance systems.

### **5.2.2 Advanced Deep Learning Models**

More advanced deep learning architectures such as CNN-LSTM, Transformer models, and attention-based networks can be used for improved temporal motion analysis and posture recognition.

### **5.2.3 Expanded Dataset Collection**

Increasing the dataset size with more therapists, body types, age groups, and environmental conditions can improve model generalization and robustness.

### **5.2.4 Integration of Additional Sensors**

Additional healthcare sensors such as EEG sensors, EMG sensors, and pressure mapping sensors can be integrated for more detailed physiological monitoring.

### **5.2.5 Mobile Application Development**

A dedicated mobile application can be developed for remote monitoring, therapy tracking, and personalized healthcare recommendations.

### **5.2.6 Clinical Validation**

Future studies can include clinical trials and medical validation to correlate posture quality and motion parameters with therapeutic outcomes.

### **5.2.7 Cloud-Based Analytics**

Advanced cloud analytics and Internet of Medical Things (IoMT) integration can be implemented for large-scale healthcare data analysis and remote therapy management.



### 5.2.8 Standardization of Ayurvedic Therapy Evaluation

The proposed framework can contribute toward creating standardized computational metrics for evaluating traditional Ayurvedic therapies and therapist performance.

The future improvements can further enhance the accuracy, scalability, and real-world applicability of the proposed intelligent Shiro Abhyanga evaluation system.

## Chapter 6

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